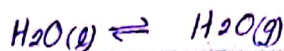
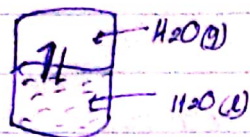


Chapter 02:- Phase Equilibrium



This is called a dynamic equation.

If we had ice it will be, $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$

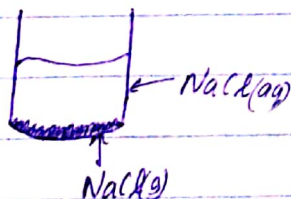
Gibbs Phase Rule

The Gibbs phase rule states that the provided equilibrium between any number of phases is only influenced by temperature, pressure and concentration.

The number of degrees of freedom (F) of the system is related to the number of components (C), number of phases (P) by the Gibbs phase rule.

$$F = C - P + 2$$

Phase (P)



Phase is defined as any homogeneous physically distinct part of a system which is separated from other such part of the system by a defined boundary surface.

The smallest number of independent variables consequently taking part in state of equilibrium by means of which the composition of each phase can be expressed directly or in the form of chemical equations.

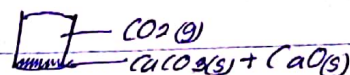
$$C = N - E$$

C; number of components

N; number of species present

E; number of independent chemical equations, relative to composition or concentration.

ex:-



$\text{CaO}(s)$ and $\text{CaCO}_3(s)$ are two different distinct phases.

This have 3 consequences (number of species present)

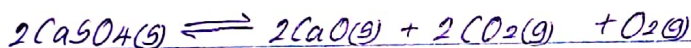
Only one equation. (decomposition of CaCO_3)

$$C = N - E$$

$$= 3 - 1$$

$$= 2.$$

To define composition of each phase we need only two components.



Gasses can mix in all proportion to form one phase

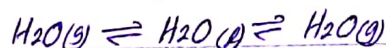
$$P = 3$$

$$C = 4 - 1 = 3$$

★ Homogeneous systems are known as 1 phase system.

• Component; minimum number of independent species necessary to define the composition of all phases of the system.

One component system, $C = 1$



The degree of freedom.

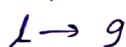
* the number of intensive variables that are independent of each other, ~~or in a~~

The degree of freedom defined as the minimal number of variables that can be change independently without describing the number of phases in equilibrium.

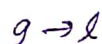
Phase diagram.

Phase diagram is a graphical representation of variase phases of a substance or mixture of substances, co exist in thermodynamic equilibrium.

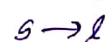
Vaporization



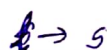
Condensation



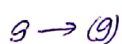
Melting



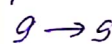
Freezing



Sublimation

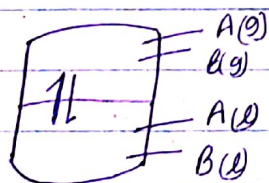


Deposition



* extra (A/L)

Racal Laco.



} saturated system.

$$P_A = X_A P_A^\circ$$

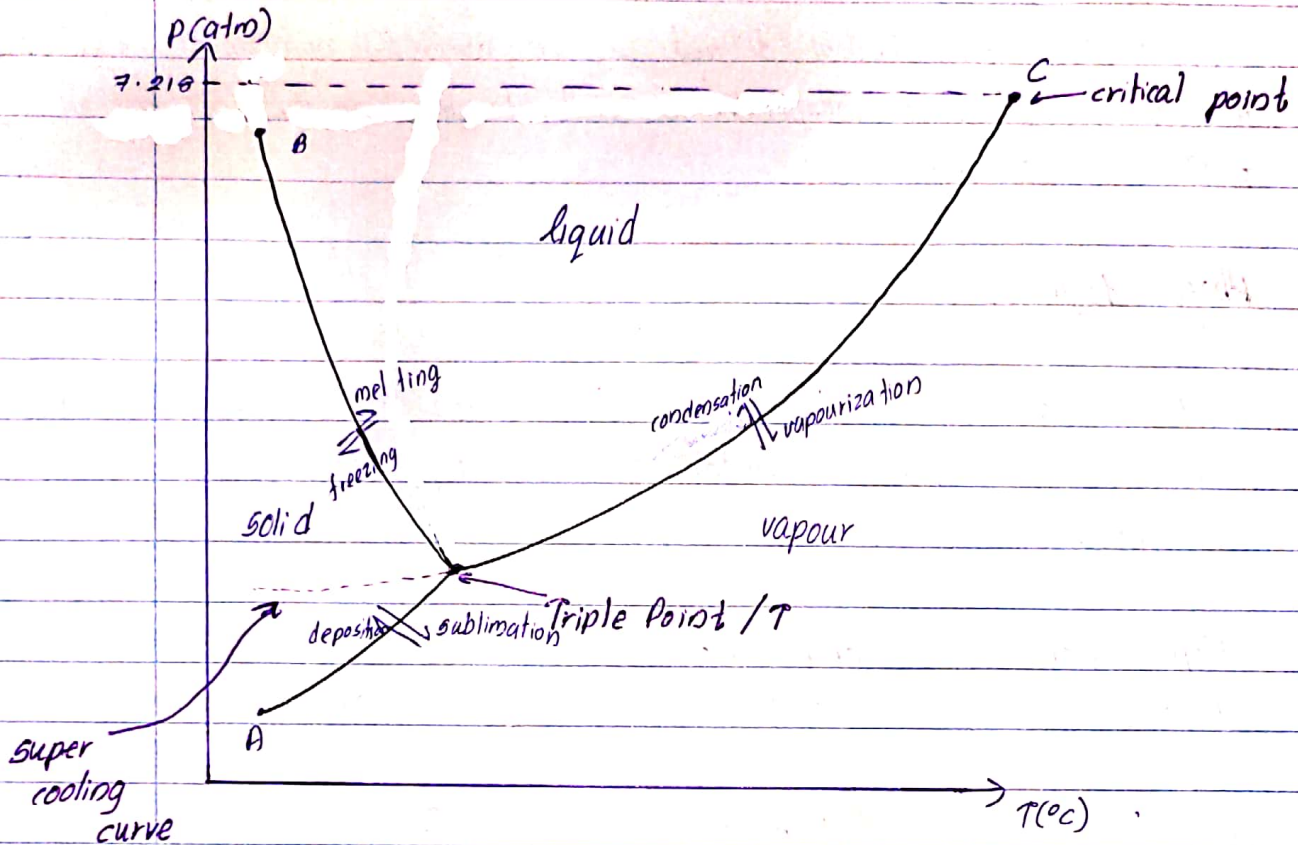
$$P_B = X_B P_B^\circ$$

X_A - mol fraction of ~~the~~ A (g)

RICHARD

Phase diagram

Phase diagram is a graphical representation of various phases of a substance or mixture of substances, coexist in the



In point in any line shows the equilibrium between two phases.
In a point there phases are at equilibrium (triple point)

* two distinguishable phases; Two phases are considered distinguishable when they have distinct properties

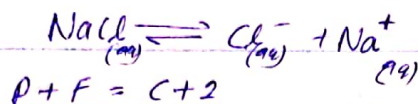
density - liquid is much denser than gas

compressibility - gas is more compressible than liquid.

appearance - a clear surface separates them.

Above the critical temperature, substance enters a state called as supercritical fluid. substance has both properties of liquid and can flow like a gas and dissolve like a substance.
two phases are distinguishable

Supercooling refers to the process of cooling a liquid below its freezing point. In super cooling curve, the temperature can be low below the normal freezing point without appearing the solid. In higher temperature above gas can't be liquified.



$$P + F = C + 2$$

$$1 + F = 1 + 2$$

$$F = 2.$$

When $F = 2$, it is a bivariant system.

Can independently change two variables (such as temp & press) while maintaining the equilibrium phases.

When $F = 0$, it is a invariant system can't change any variables.

Chemical Potential (μ)

Chemical substances move from higher to lower chemical potential by diffusional transport and chemical reactions.

The driving force in a physico chemical system at constant temperature and pressure is the gradient of chemical potential.

$$G = f(T, P, n_2)$$

$$dG = \left(\frac{\partial G}{\partial T} \right)_{P, n_2} dT + \left(\frac{\partial G}{\partial P} \right)_{T, n_2} dP + \left(\frac{\partial G}{\partial n_2} \right)_{P, T} dn_2$$

For two substances A and B.

$$\frac{dG}{dT} = \left(\frac{\partial G}{\partial T} \right)_{P, n_A, n_B} dT + \left(\frac{\partial G}{\partial P} \right)_{T, n_A, n_B} dP + \left(\frac{\partial G}{\partial n_A} \right)_{P, T, n_B} dn_A + \left(\frac{\partial G}{\partial n_B} \right)_{P, T, n_A} dn_B$$

$$\left(\frac{\partial G}{\partial T}\right)_P = -S$$

$$\left(\frac{\partial G}{\partial P}\right)_T = +V$$

$$dG = VdP - SdT + \sum \left(\frac{\partial G}{\partial n_i}\right)_{P,T} dn_i$$

at constant Temp. and pressure

$$dG = \sum \left(\frac{\partial G}{\partial n_i}\right)_{P,T} dn_i$$

$$\left(\frac{\partial G}{\partial n_i}\right)_{P,T} = \mu_i$$

$$dG = \sum \mu_i dn_i$$



$$\mu_{H_2O}^L < \mu_{H_2O}^S < \mu_{H_2O}^g$$

lowest chemical potential $\rightarrow$ stable phase.

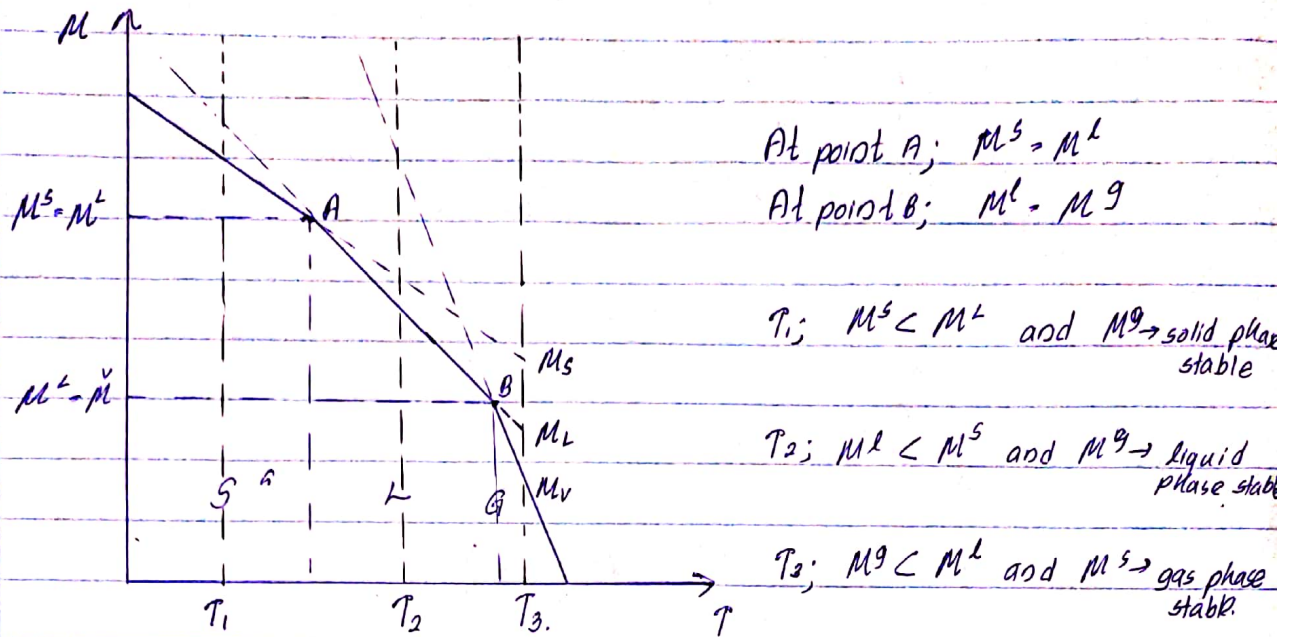
$$\left(\frac{\partial \mu}{\partial T}\right)_P = \bar{s}$$

molar entropy

$$\left(\frac{\partial \mu}{\partial P}\right)_T = \bar{v}$$

molar volume.

Change of Chemical Potential with Temperature



As the pressure increases boiling point and melting point are increasing.



$$A \rightleftharpoons B$$

at equilibrium $M_A = M_B$

$$\frac{dG}{n} = \frac{-SdT}{n} + \frac{VdP}{n}$$

$$dM \Rightarrow \bar{S}dT + \bar{V}dP$$

$$-\bar{S}^A dT + \bar{V}^A dP = -\bar{S}^B dT + \bar{V}^B dP$$

$$(\bar{S}^A - \bar{S}^B) dT = (\bar{V}^A - \bar{V}^B) dP$$

$$\Delta \bar{S} dT = \Delta \bar{V} dP$$

$$\boxed{\frac{dP}{dT} = \frac{\Delta \bar{S}}{\Delta \bar{V}}}$$

← clapyron equation

$$\Delta \bar{S} = \left(\frac{dG}{dT} \right)_P = \left(\frac{\Delta H}{T} \right)$$

Solid Liquid Equilibrium $S \rightleftharpoons L$

$$\frac{dP}{dT} = \frac{\Delta H}{T \Delta V}$$

$$\Delta \bar{H} = \Delta \bar{H}^0 \quad (\text{we assume that } \Delta H \text{ is independent})$$

$$\frac{dP}{dT} = \frac{\Delta \bar{H}^0}{\Delta V} \frac{1}{T}$$

$$\int_{P_1}^{P_2} dP = \frac{\Delta \bar{H}^0}{\Delta V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

$$P_2 - P_1 = \frac{\Delta \bar{H}^0}{\Delta V} \ln \left(\frac{T_2}{T_1} \right)$$

$$\Delta P = \frac{\Delta \bar{H}^0}{\Delta V} \ln \left(1 + \frac{T_2 - T_1 + T_1}{T_1} \right)$$

$$\Delta P = \frac{\Delta \bar{H}^0}{\Delta V} \ln \left(1 + \frac{\Delta T}{T_1} \right)$$

$$\Delta P = \frac{\Delta \bar{H}^0}{\Delta V} \left(\frac{T_2 - T_1}{T_1} \right)$$

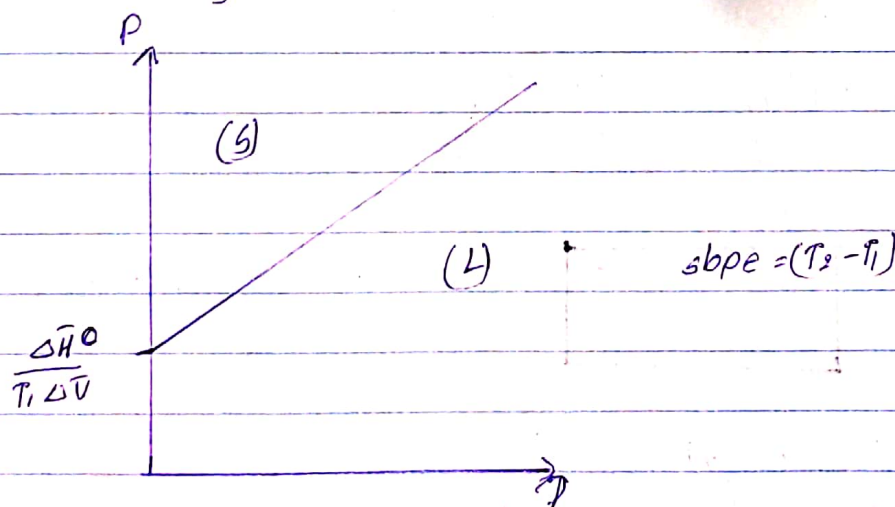
$T_2 - T_1$ is small

so let $\frac{\Delta T + T_1}{T_1} = x$

$$\therefore \ln(1+x) = x \quad x \rightarrow 0$$

$$P_2 = \frac{\Delta \bar{H}^0}{T_1 \Delta V} (T_2 - T_1) + P_1$$

$\downarrow$ $\downarrow$ $\downarrow$
 y m x $+ c$



Liquid Vapour Equilibrium $L \rightleftharpoons V$

$$\frac{dP}{dT} = \frac{\Delta \bar{S}}{\Delta \bar{V}}$$

$\Delta \bar{V} = \bar{V}_L - \bar{V}_g$ (since gases have bigger volume we can assume $\bar{V}_g \gg \bar{V}_L$)
 $\Delta \bar{V} = \bar{V}_g$

$$\frac{dP}{dT} = \frac{\Delta \bar{H}^\theta}{T \bar{V}_g} \quad \left(\Delta \bar{S} = \frac{\Delta \bar{H}}{T} \right)$$

$$\bar{V}_g = \frac{RT}{P}$$

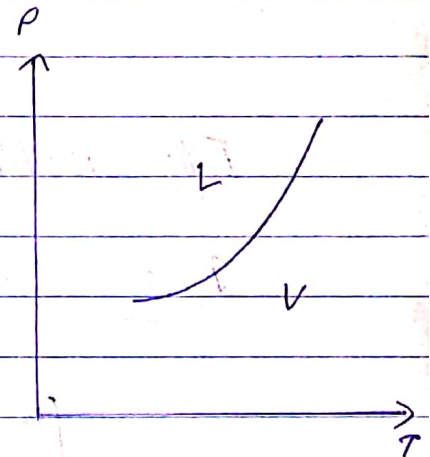
$$\frac{dP}{dT} = \frac{\Delta \bar{H}^\theta}{RT^2} P$$

$$\frac{dP}{dT} = \frac{\Delta \bar{H}^\theta}{RT^2} P$$

$$\int_{P_1}^{P_2} \frac{1}{P} dP = \frac{\Delta \bar{H}^\theta}{R} \int_{T_1}^{T_2} \frac{dT}{T^2}$$

$$\ln(P_2/P_1) = \frac{\Delta \bar{H}^\theta}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$P_2 = P_1 e^{-\frac{\Delta \bar{H}^\theta}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$



This gives an exponential curve.

Two component system.

$$P + F = C + 2$$

$$P + F = 4$$

when $P = 2$ $F = 2$

when $P = 1$ $\therefore F = 3$ $\leftarrow$ composition.

RICHARD

Solid / Liquid.

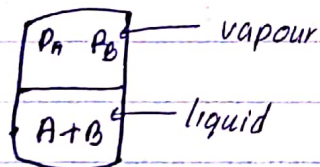
Solid / Solid

Solid + Liquid / Liquid.

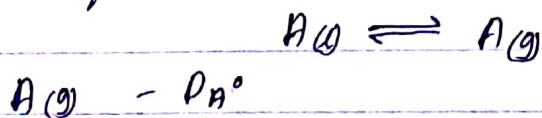
- Two liquids fully visible. Can mix in all proportion.
benzene / toluene - ideal solutions.
nitrobenzene / benzene - non ideal.
- Partially visible
phenol / H₂O
- Immisible
CCl₄ / H₂O

Fully misible liquid system

Ideal System



When a liquid at a equilibrium with it's vapour the pressure exerted by the vapour is known as the vapour pressure of the liquid.



* dynamic equilibrium means a equilibrium that dosen't stop.

Roult's law

$$P_A \propto X_A$$

$$P_A = P_A^\circ X_A$$

$$P_B = P_B^\circ X_B$$

P_A ; Partial pressure of A

X_A ; Mol fraction of A in the liquid phase

By Dalton's law of partial pressure,

$$P_T = P_A + P_B$$

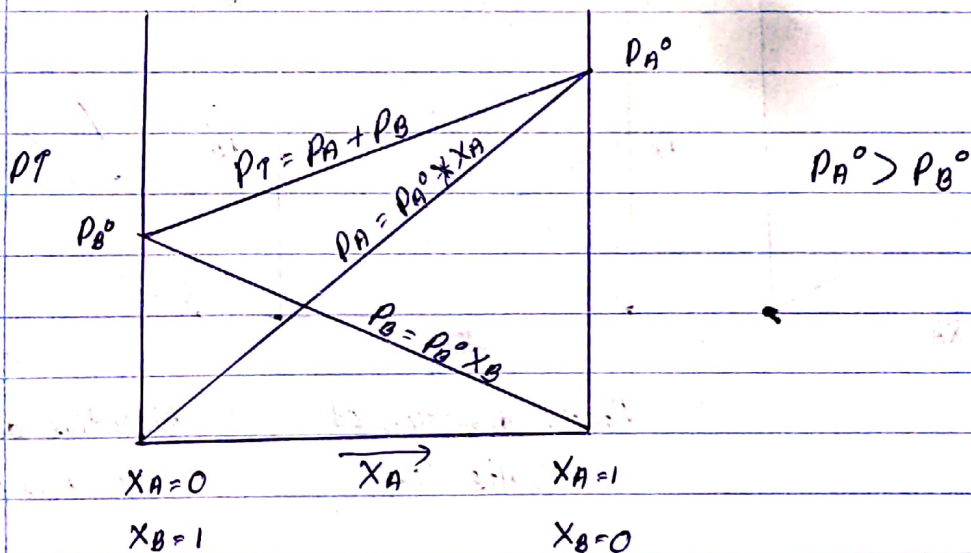
$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$P_T = P_A^\circ X_A + P_B^\circ (1 - X_A)$$

$$P_T = P_B^\circ + (P_A^\circ - P_B^\circ) X_A$$

$$P_T = (P_A^\circ - P_B^\circ) X_A + P_B^\circ$$

$$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow$$
$$y = \quad m \quad x + c$$



For an ideal solution = A-A = B-B = A-B

So no changing upon mixing.

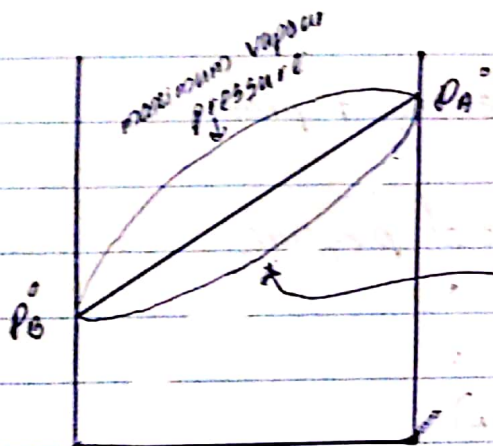
RICHARD

Ideal solution behave according to the Raoult's law.

$$A-B < A-A \text{ or } B-B$$

Because of this, the mixture becomes more volatile.
That's why water has higher boiling point than methanol.

Positive deviation from the Raoult's law,

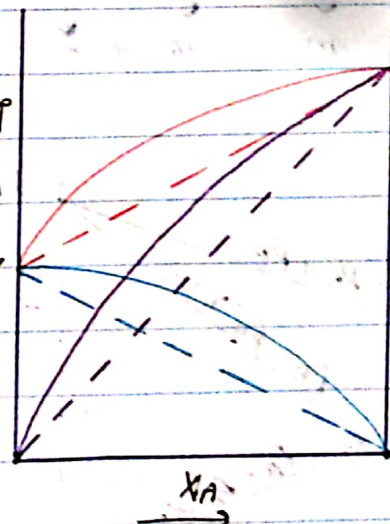
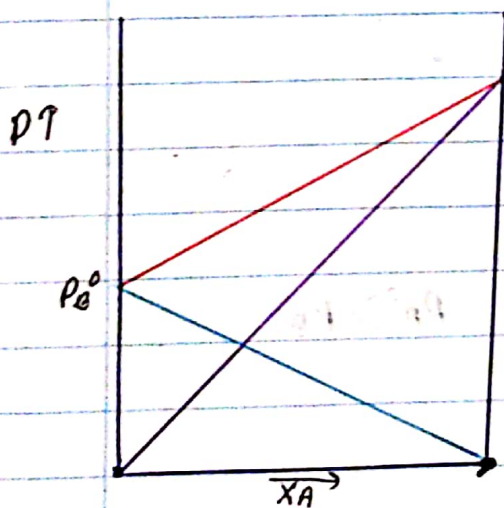


In here,

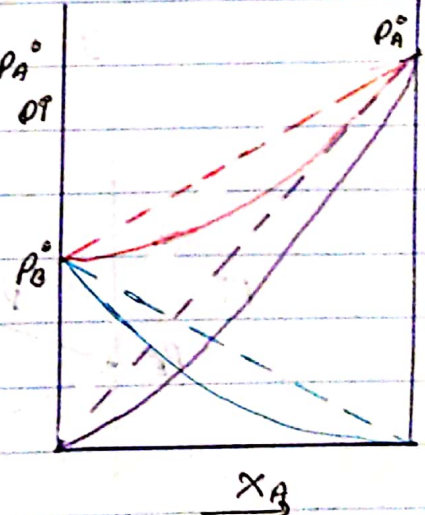
$$B-A \gg B-B > A-A$$

Mixture has higher boiling point.
So this is less volatile.

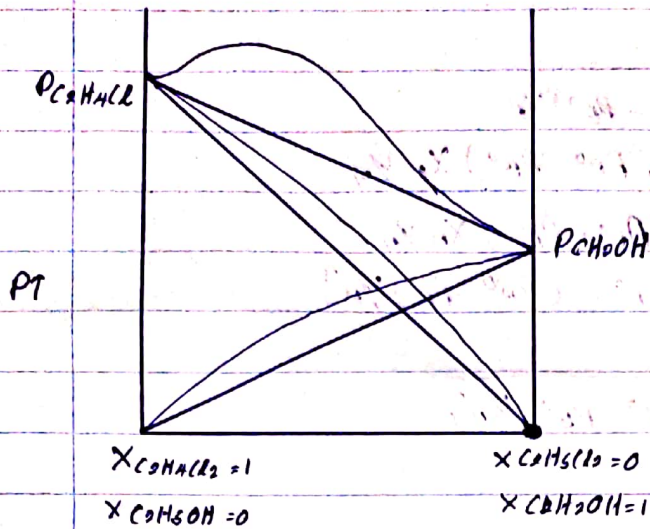
extra,



Positive deviation of
Raoult's law



Negative deviation
of Raoult's law.



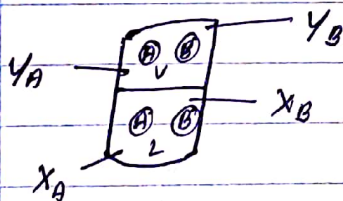
The experimentally measured pressure for all compositions of the mixture are higher than the calculated order into Raoult's law. Therefore the solution is not ideal. And deviate from the Raoult's law in the positive direction.

Completely miscible liquid liquid system.

Absence of a minimum or maximum in the vapour pressure curve (heterotropic)

The presence of maximum or minimum in the vapour pressure curve (H₂O / EtOH)

Variation of vapour pressure with the composition of the vapour phase.



$$P_A^0 > P_B^0$$

A is more volatile. Have a less boiling point

$$P_A = P Y_A$$

$$Y_A = \frac{P_A}{P}$$

$$P_B = P Y_B$$

$$Y_B = \frac{P_B}{P}$$

$$X_B = 1 - X_A$$

$$P_T = P_A + P_B$$

$$P_A = X_A P_A^\circ \quad \text{--- (1)}$$

$$P_B = X_B P_B^\circ \quad \text{--- (2)}$$

From (1) and (2),

$$P_T = X_A P_A^\circ + X_B P_B^\circ$$

$$P_T = X_A P_A^\circ + (1 - X_A) P_B^\circ$$

$$P_T = X_A P_A^\circ - X_A P_B^\circ + P_B^\circ$$

$$P_T = (P_A^\circ - P_B^\circ) X_A + P_B^\circ \quad \text{--- (3)}$$

$$Y_A = \frac{P_A}{P_T}$$

From (1) and (3)

$$Y_A = \frac{X_A P_A^\circ}{(P_A^\circ - P_B^\circ) X_A + P_B^\circ}$$

$$X_A P_A^\circ = P_B^\circ Y_A + (P_A^\circ - P_B^\circ) Y_A X_A$$

$$X_A P_A^\circ = P_B^\circ Y_A + P_A^\circ Y_A X_A - P_B^\circ Y_A X_A$$

$$P_B^\circ Y_A - X_A P_A^\circ = P_A^\circ Y_A X_A - P_B^\circ Y_A X_A$$

$$P_B^\circ Y_A = X_A (P_A^\circ - P_A^\circ Y_A + P_B^\circ Y_A)$$

$$P_B^\circ Y_A = X_A \{ P_A^\circ - (P_A^\circ - P_B^\circ) Y_A \}$$

$$X_A = \frac{P_B^\circ Y_A}{P_A^\circ - (P_A^\circ - P_B^\circ) Y_A} \quad \text{--- (4)}$$

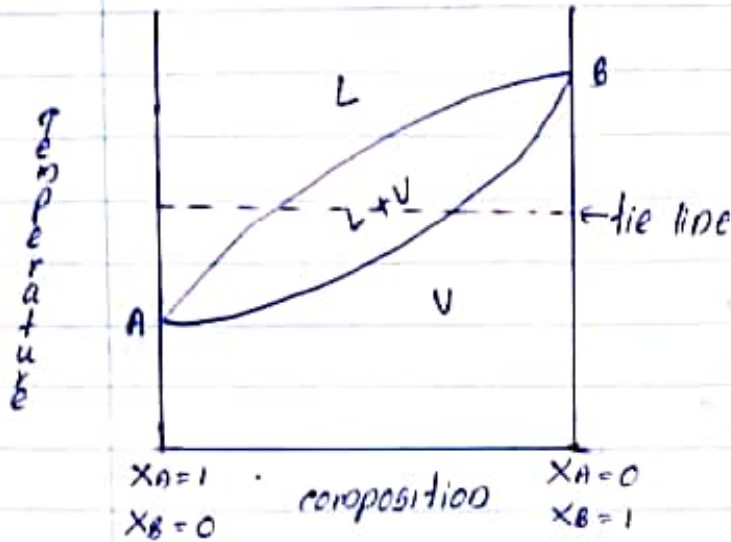
From (4) and (3) (substituting (4) X_A to (3))

$$P_T = P_B^\circ + (P_A^\circ - P_B^\circ) \times \frac{P_B^\circ Y_A}{P_A^\circ - (P_A^\circ - P_B^\circ) Y_A}$$

$$= \frac{P_B^\circ P_A^\circ - P_A^\circ P_B^\circ Y_A + P_B^{2\circ} Y_A + P_A^\circ P_B^\circ Y_A - P_B^{2\circ} Y_A}{P_A^\circ - (P_A^\circ - P_B^\circ) Y_A}$$

$$= \frac{P_A^\circ P_B^\circ}{P_A^\circ - (P_A^\circ - P_B^\circ) Y_A}$$

$$P_A = \frac{P_A^\circ P_B^\circ}{P_A^\circ + (P_B^\circ - P_A^\circ) Y_A}$$



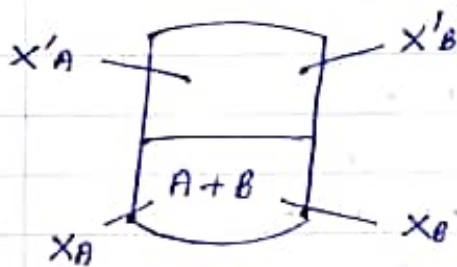
When vapour pressure is high
boiling point is low.

When vapour pressure is low
boiling point is high

$$\text{Vapour pressure} \propto \frac{1}{\text{boiling point}}$$

Temperature Vs Composition

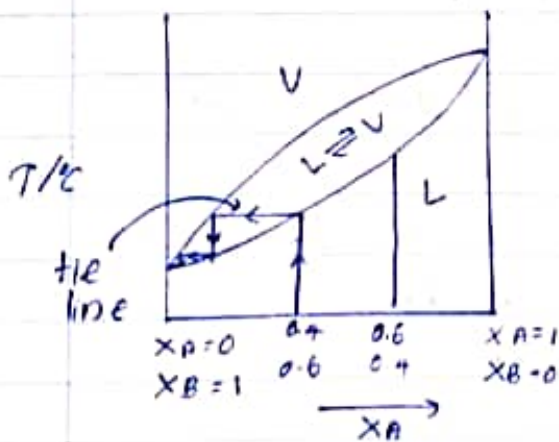
Completely miscible liquids.



X' - molfraction of vapour phase
 X - molfraction of liquid phase

If boiling point decreases then
vapour pressure gets high.

Phase diagram temperature composition



tie line; represent horizontal line
separate ~~that~~ draw to
vapour phase from the
liquid phase.

equimolar mixture; same number of
moles for all compounds

The boiling points of compound are changing due to the
the composition.

$$P_A^{\circ} = 60 \text{ Hgmm}$$

$$P_B^{\circ} = 90 \text{ Hgmm}$$

$$P_A = P_A^{\circ} X_A$$

$$= 60 \times \frac{1}{2}$$

$$= 30$$

$$P_B = P_B^{\circ} X_B$$

$$= 90 \times \frac{1}{2}$$

$$= 45$$

$$P_T = P_A + P_B = 30 + 45 = 75 \text{ Hgmm}$$

$$P_A = X'_A P_T$$

$$X'_A = \frac{P_A}{P_T}$$

$$= \frac{30}{75}$$

$$= 0.4$$

$$P_B = X'_B P_T$$

$$X'_B = \frac{P_B}{P_T}$$

$$= \frac{45}{75}$$

$$= 0.6$$

$$P_A = P_A^{\circ} X_A$$

$$= 60 \times 0.4$$

$$= 24 \text{ Hgmm}$$

$$P_B = P_B^{\circ} X_B$$

$$= 90 \times 0.6$$

$$= 54 \text{ Hgmm}$$

$$P_T = P_A + P_B = 24 + 54 = 66 \text{ Hgmm}$$

$$P_A = X'_A P_T$$

$$X'_A = \frac{24}{66} = 0.3$$

$$P_B = X'_B P_T$$

$$X'_B = \frac{54}{66} = 0.7$$

The calculation shows the vapourization followed by condensation. the vapour could be much richer more volatile B.

By the liquid richer in less volatile A.

The vapour at T_1 and subsequently will give per liquid B.

Fractional distillation

Fractional distillation is a technique used to separate a mixture of liquids into its individual components based on their different boiling points. It involves the use of a spiral bubble plate column, where the mixture is heated and then vaporized.



As the vapour rises through the column, it condenses at different levels depending on the boiling points. The more volatile components with lower boiling points will condense at the top of the column.

The process works in according to the Raoult's law. If solution not works under it we never get 100% from distillation.

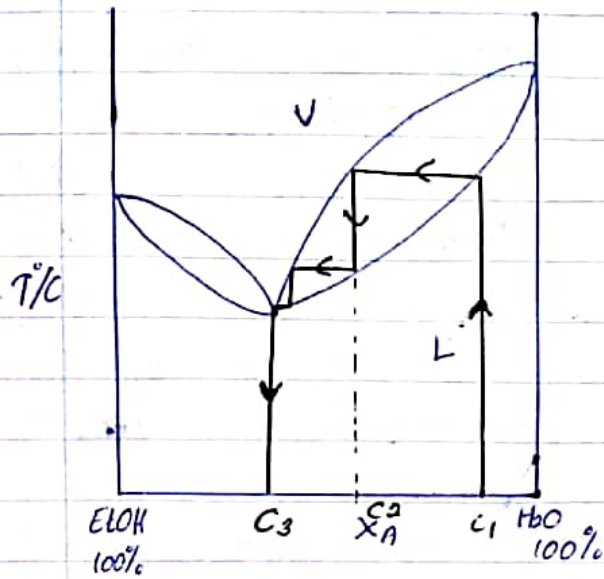
In the case of crude oil, fractional distillation is used to separate various products such as petrol and diesel.

These products are separated based on their molecular weights. Lighter molecules tend to have lower boiling points. Heavier molecules tend to have higher boiling points.

Azeotropes

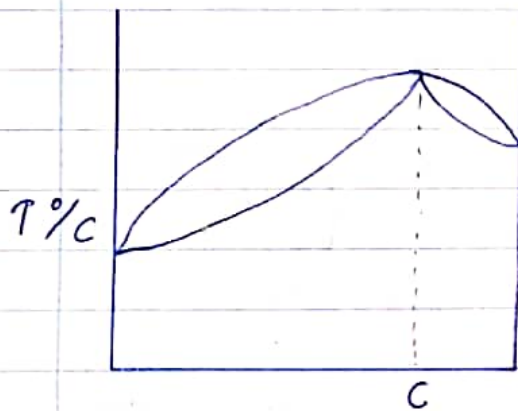
The liquid mixture that can't be separated by fractional distillation. Because the composition of the vapour coming off the liquid is the same as the liquid itself. This azeotropic behaviour is said to result from deviation from ideality. Especially deviation from the Raoult's law.

Minimum boiling azeotropic mixture, $\text{CH}_3\text{O} / \text{C}_2\text{H}_5\text{OH}$ 4% 96%



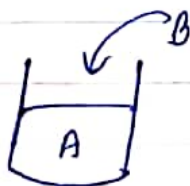
The vapour phase become more and more richer in EtOH, and the liquid phase becomes richer in water. After several distillation steps the liquid mixture reaches to your minimum boiling point at this composition. C3. both EtOH and H2O distilled as a single liquid

Maximum boiling azeotropes $\text{C}_2\text{H}_5\text{Cl}_3 / \text{Acetone}$, $\text{H}_2\text{O} / \text{HCl}$ 52% 48%

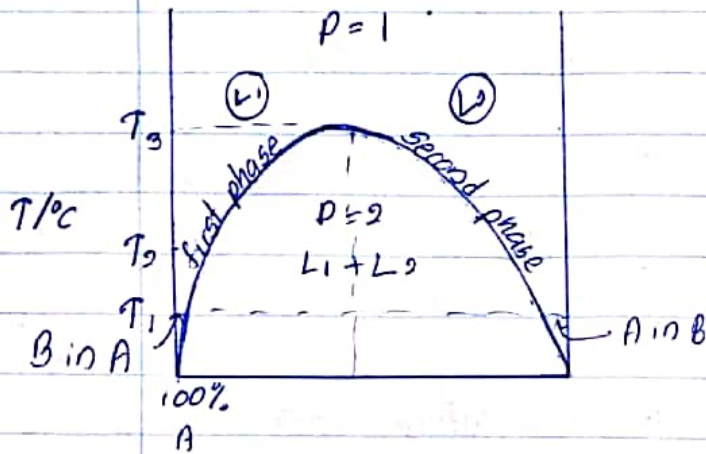


Partially miscible liquid mixture

$\text{PhOH} / \text{H}_2\text{O}$
 $\text{ether} / \text{H}_2\text{O}$



$$A-B < \begin{matrix} A-A \\ B-B \end{matrix}$$



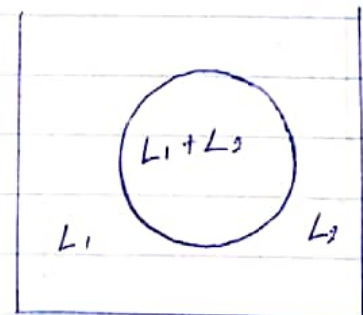
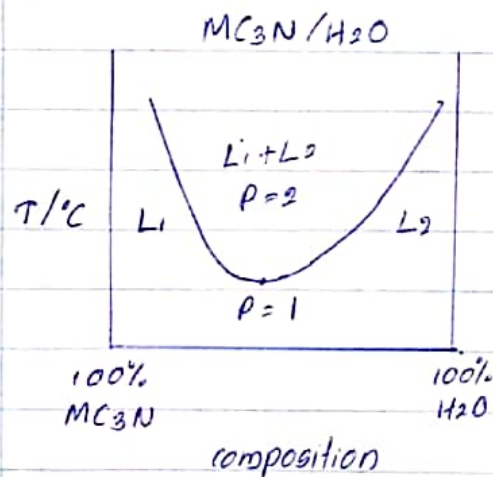
A small amount of liquid B is added at T_1 . It dissolves completely in the beginning forming only one phase.

After that the liquid B doesn't dissolve but forms another layer. The system now consists of two phases.

After that again one phase is obtained. In the region A and B two phases called conjugate solution.

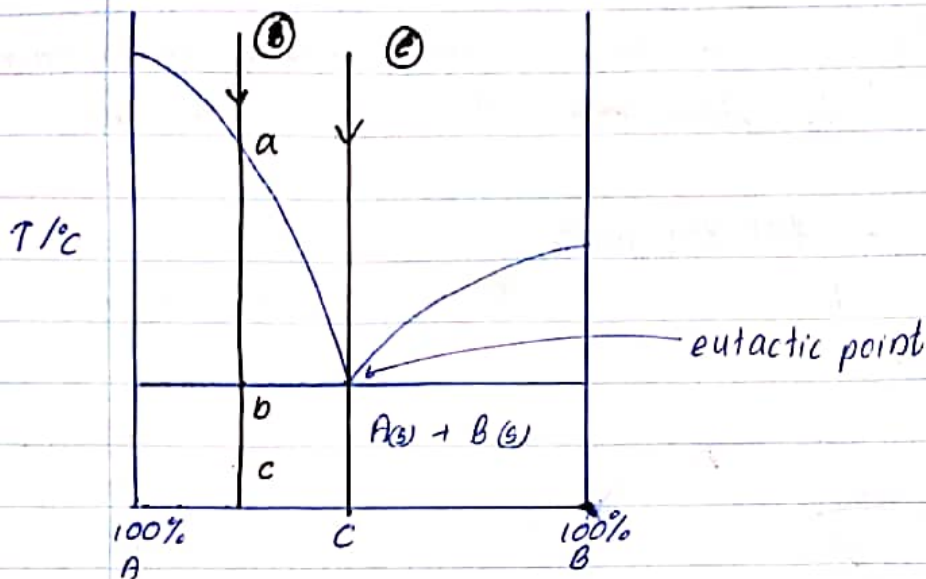
The composition of the two phases at equilibrium varies with the temperature. The rising the temp. miscibility increases.

Above the temperature T_3 , the liquids are completely miscible.

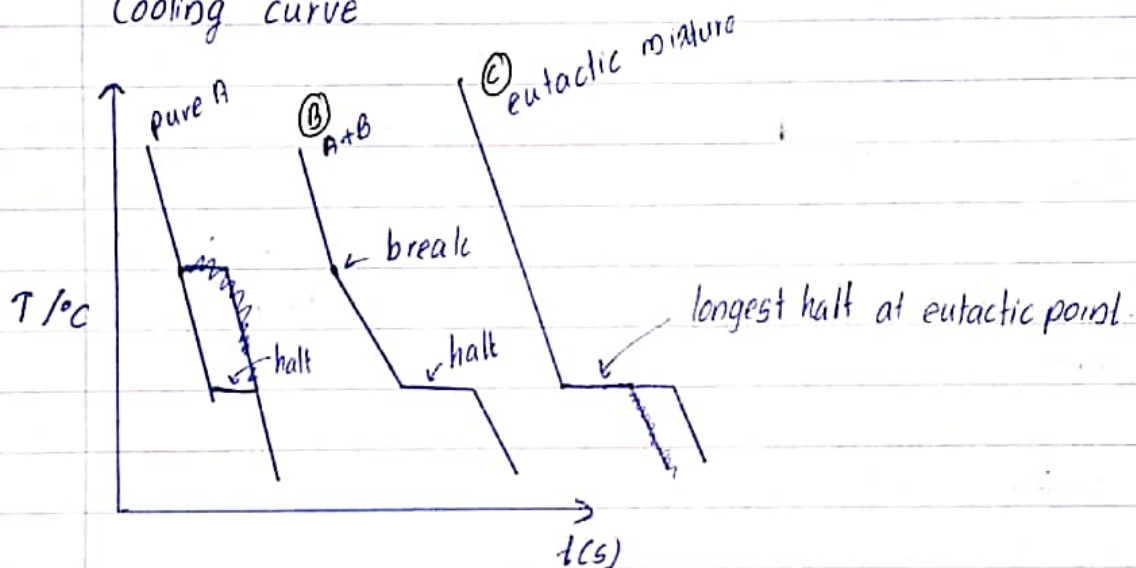


Nicotine/water.

They form a weak complex.



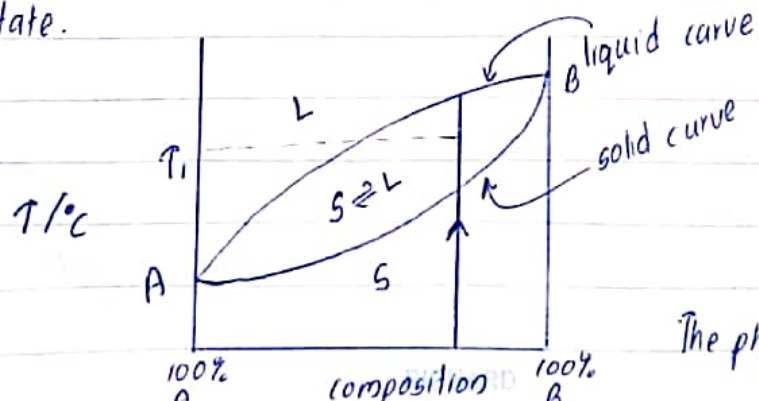
Cooling curve



lowest melting point get at the eutactic point.

Alloys

Two metals which are completely soluble in liquid state as well as solid state.



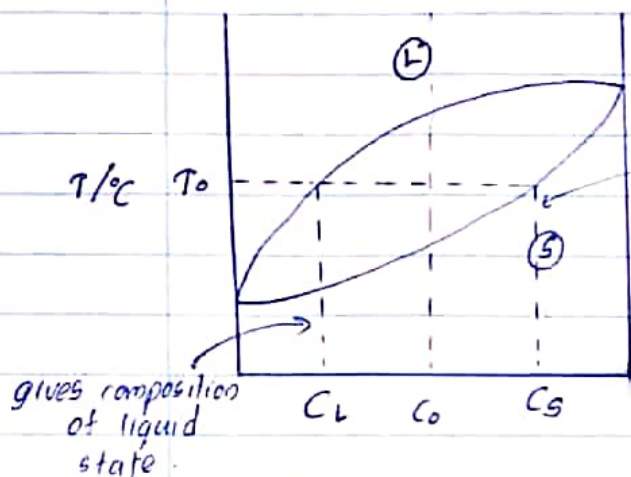
solid form - completely miscible.

The phase diagram looks like a liquid mixture.

To determine the actual composition of the phases in equilibrium at any temperature in the two phase region, draw a horizontal line. (tie line)

The relative amount of different phases.

We apply lever's rule to determine the relative amount of different phases at equilibrium.



This gives the composition of solid state.

W_s ; mass fraction of solid state.

W_L ; mass fraction of liquid state.

$$W_s + W_L = 1 \quad \text{--- (1)}$$

$$W_s = 1 - W_L$$

$$W_s C_s + W_L C_L = C_0 \quad \text{--- (2)}$$

$$(1 - W_L) C_s + W_L C_L = C_0$$

$$C_s - W_L C_s + W_L C_L = C_0$$

$$W_L C_L - C_0 + C_s = C_s W_L = 0$$

$$W_L (C_L - C_s) = C_0 - C_s$$

$$W_L = \frac{C_0 - C_s}{C_L - C_s}$$

like wise

$$W_s = \frac{C_0 - C_L}{C_s - C_L}$$

$$\left. \begin{array}{l} C_0 ; 35\% \text{ Ni} \\ C_L ; 31.5\% \text{ Ni} \\ C_s ; 42.5\% \text{ Ni} \end{array} \right\} \text{ At } T_1 \text{ Temperature.}$$

What is the mass fraction of Nickel in the solid and liquid phase.

$$W_L = \frac{C_0 - C_S}{C_L - C_S}$$

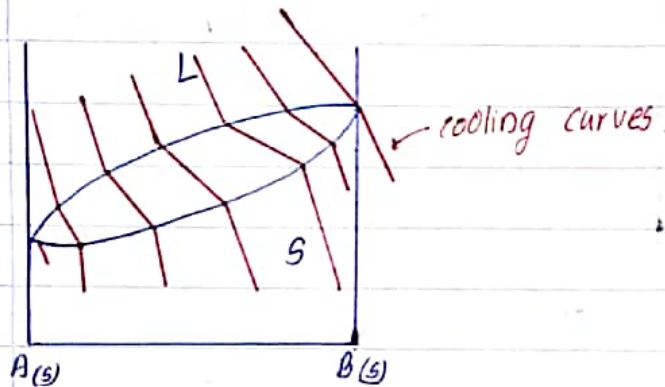
$$= \frac{35 - 42.5}{31.5 - 42.5}$$

$$= 0.651615$$

$$W_S = \frac{C_0 - C_L}{C_S - C_L}$$

$$= \frac{35 - 31.5}{42.5 - 31.5}$$

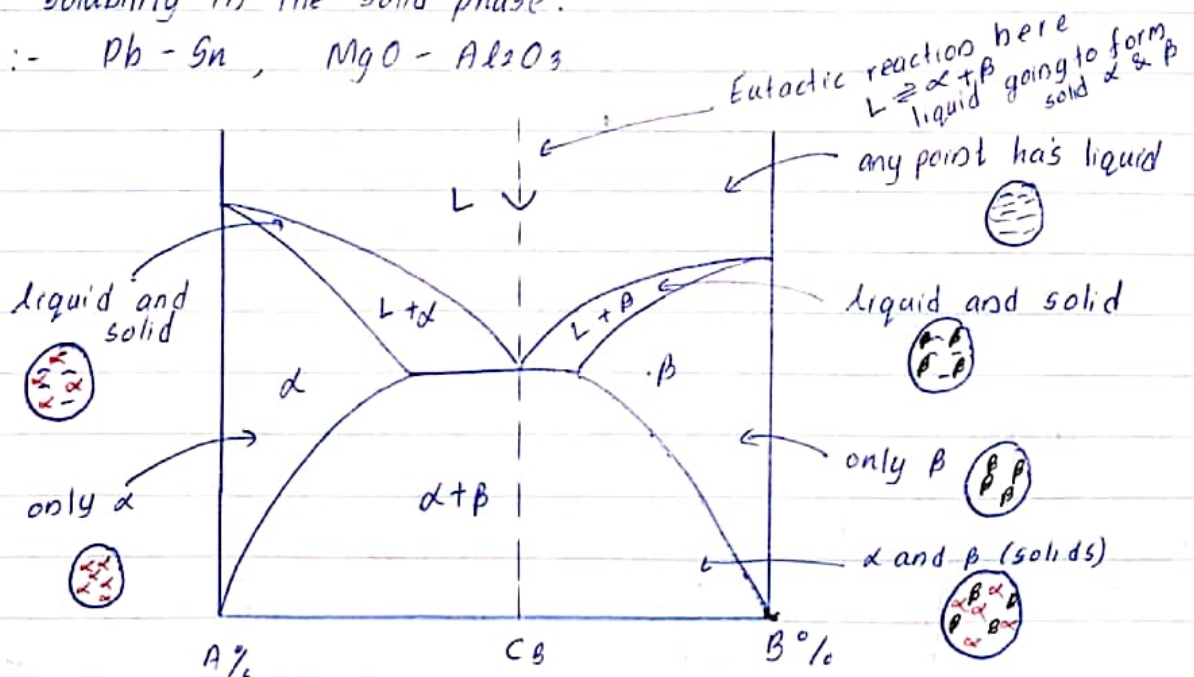
$$= 0.34545$$



Binary Eutactic Phase Diagram

Two components having complete solubility in liquid phase, but partial solubility in the solid phase.

ex:- Pb - Sn, MgO - Al₂O₃



$$F = C - P + 1$$

$$= 2 - P + 1$$

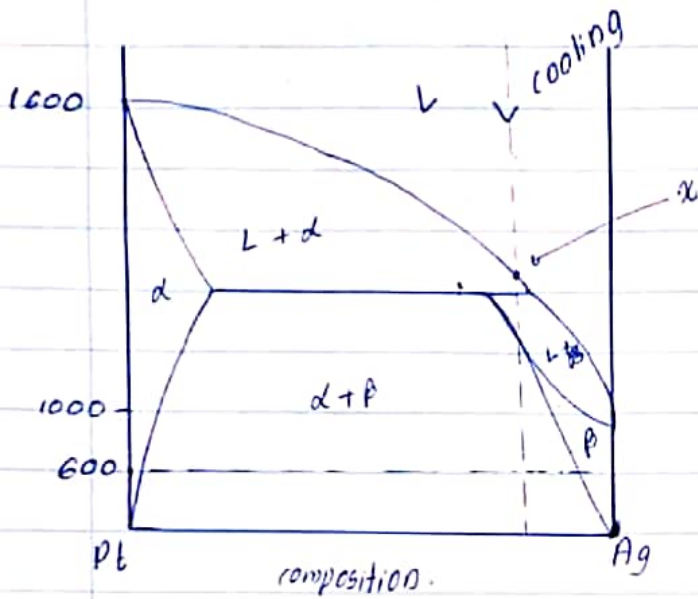
$$= 3 - P$$

pressure constant

at liquid phase $F = 2$ (because one phase)
at eutactic point $F = 0$

RICHARD

Peritectic Reaction

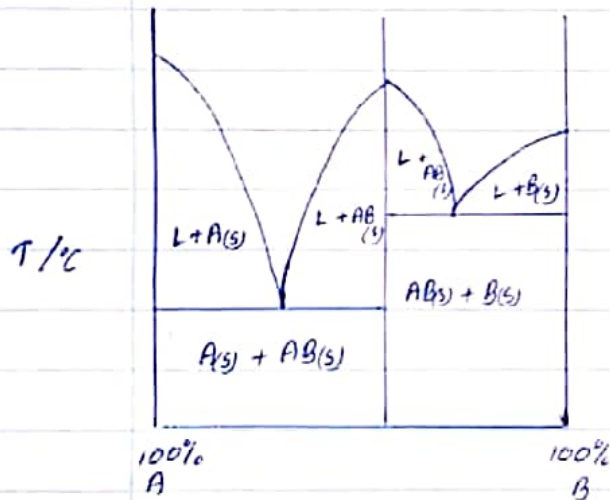


At a liquid α starts to solidification

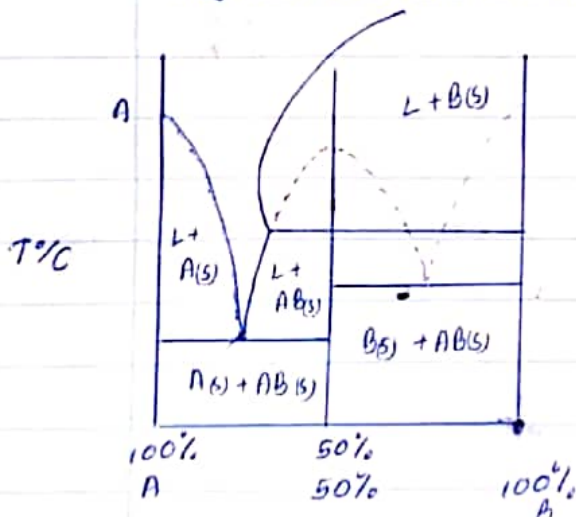


The solid react with the liquid giving another solid phase.

This is called peritectic reaction.



congruent formation of $AB(s)$



AB very unstable.

$AB(s)$ can't exist with the liquid in the equilibrium.

If it's stable we can see the phase diagram as in the dotted line.